RENAL PATHWAYS — CLEAN PARALLEL MAP

Read one row LEFT → RIGHT. Repeated structures are the SAME structures. No lines connect different rows or panels.

Solid arrows = forward pathway. Red dashed local return = negative feedback. A–H group related pathways; row IDs replace long cross-links.

Important structures / receptors / proteins / sites retained. Intracellular second messengers and phosphorylation sites omitted.

A OSMOTIC SENSING & VP SOURCE

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

A1 Osmotic input via SFO / OVLT

Plasma osmolality ↑
Dehydration / small hypertonic-saline load
→ **SFO / OVLT**
Cell shrinkage → TRPV-related cation entry / firing ↑
→ **MnPO**
Relays osmotic information
→ **Magnocellular PVN / SON**
VP synthesised here
Firing ↑; axons to posterior pituitary terminals
→ **Posterior-pituitary**
axon terminals
VP released into blood ↑

VP production and release are different sites. Renal actions continue in B1–B2.

A2 Intrinsic osmosensitivity of VP neurons

Plasma osmolality ↑
→ **PVN / SON neurons**
Water exits; cells shrink
→ **Mechanosensitive**
cation conductance ↑
Depolarisation / firing ↑
→ **Posterior-pituitary**
axon terminals
Contain stored VP
→ **VP release into blood ↑**

PVN / SON can sense osmotic change as well as receive SFO / OVLT input.

A3 Osmotic thirst

Plasma osmolality ↑
→ **SFO / OVLT**
Osmotic sensing
→ **MnPO → cortical**
thirst circuits
→ **Thirst / drinking ↑**
→ **New water intake ↑**
Osmolality toward normal

NEGATIVE FEEDBACK: Osmolality restored → osmotic thirst stimulus ↓.

A4 Low osmolality: reverse water response

Plasma osmolality ↓
Excess water
→ **SFO / OVLT + PVN / SON**
Cell swelling → cation conductance / firing ↓
→ **Posterior-pituitary**
VP release ↓
→ **Collecting-duct V_2 drive ↓**
APICAL AQP2 availability ↓
→ **More dilute urine ↑**
Excess water removed
Osmolality restored

NEGATIVE FEEDBACK: Removal of excess water resolves the low-osmolality stimulus.

B VP EFFECTS: RECEPTORS & TARGETS

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

B1 VP → collecting-duct water reabsorption

Circulating VP ↑
From A1 / A2 / C / E3
→ **Collecting-duct principal cell**
BASOLATERAL V_2 receptor
→ **APICAL AQP2 insertion ↑**
Water enters from lumen
→ **BASOLATERAL AQP3 / AQP4**
Water exits to interstitium → blood
→ **Water reabsorption ↑**
Urine volume ↓
Osmolality toward normal

NEGATIVE FEEDBACK: Corrected osmolality reduces osmotic VP drive.

Requires a functioning medullary gradient (G). AQP2 is a water channel, not a pump.

B2 VP → inner-medullary urea handling

Circulating VP ↑
→ **Inner medullary**
collecting duct (IMCD)
BASOLATERAL V_2 receptor
→ **APICAL UT-A1**
Activity / membrane availability ↑
→ **Urea: lumen → UT-A1**
→ cell → BASOLATERAL UT-A3 → interstitium
→ **Medullary urea ↑**
Gradient supported
Water extraction facilitated

UT-A1 transports urea; water extraction still requires AQP2. See G3–G4.

B3 Higher VP → direct vasoconstriction

Higher circulating VP
→ **V_1a receptors**
SYSTEMIC / CORONARY vascular smooth muscle
→ **Smooth-muscle**
contraction ↑
→ **Vasoconstriction ↑**
Peripheral resistance ↑
→ **Arterial BP supported**
toward normal

NEGATIVE FEEDBACK: Restored filling / BP reduces the depletion-related VP stimulus.

B4 VP → renin: lecture-specific model*

VP
→ **Macula-densa**
V_1a receptors*
→ **Macula densa produces**
NO + PGE_2*
→ **Afferent-arteriolar**
JG granular cells
Renin production / release ↑
→ **ANGII pathway engaged**
See D–E

* Retained as the lecture model; net renin also depends on perfusion, NaCl delivery and feedback.

C VOLUME / PRESSURE & BAROREFLEX

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

C1 Low volume → atrial receptor unloading → VP

Blood volume ↓
→ **LEFT ATRIAL filling ↓**
Stretch / receptor firing ↓
→ **Vagal afferents (CN X) ↓**
NTS ↓ → CVLM ↓
Inhibition of RVLM ↓
→ **RVLM disinhibited ↑**
Magnocellular PVN / SON ↑
Posterior-pituitary VP ↑
→ **Renal water retention ↑**
Circulating volume supported

NEGATIVE FEEDBACK: Filling restored → atrial stretch / firing recovers → non-osmotic VP drive ↓.

Brainstem routing follows the lecture's simplified integrated baroreceptor model.

C2 Low arterial BP → arterial receptor unloading → VP

Arterial BP ↓
→ **Carotid SINUS / aortic ARCH**
Stretch / receptor firing ↓
→ **Carotid sinus: CN IX**
Aortic arch: CN X
NTS ↓ → CVLM ↓
→ **RVLM inhibition ↓**
RVLM ↑ → PVN / SON ↑
Posterior-pituitary VP ↑
→ **Water conservation ↑**
Volume / BP supported

NEGATIVE FEEDBACK: BP restored → arterial firing ↑ → NTS / CVLM ↑ → RVLM / VP drive restrained.

C3 RVLM → sympathetic cardiovascular and renal outputs

RVLM activity ↑
From C1 / C2 / E5
→ **Descending spinal**
sympathetic outflow ↑
→ **Vessels: constriction ↑**
Heart: rate / contractility ↑
→ **Peripheral resistance ↑**
Cardiac output ↑
→ **Arterial BP supported**

NEGATIVE FEEDBACK: BP recovery reloads arterial receptors → RVLM / sympathetic drive ↓.

Separate renal sympathetic branch: JG β_1↑ → renin (D3).

C4 Higher filling / BP → reverse reflex

Filling / arterial BP ↑
→ **Receptor stretch / firing ↑**
NTS / CVLM ↑
RVLM inhibited
→ **Non-osmotic VP drive ↓**
Sympathetic drive ↓
Cardiac vagal drive ↑
→ **Vessels dilate; HR / CO ↓**
Renal water retention ↓
More dilute urine
→ **Excess filling / BP**
corrected toward baseline

NEGATIVE FEEDBACK: As excess filling / pressure resolves, the initiating stretch stimulus declines.

Normal BP does not imply zero VP: osmotic inputs remain active.

D RENIN TRIGGERS & ANGII FORMATION

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

D1 Renal perfusion input

Renal perfusion ↓
→ **Afferent-arteriolar**
pressure / stretch ↓
→ **JG granular cells**
Renin release ↑
→ **ANGII outputs**
Vascular + renal actions
See D4 and E
→ **Perfusion / circulating**
volume supported

NEGATIVE FEEDBACK: Restored renal perfusion reduces the JG renin stimulus.

D2 Tubular NaCl input

NaCl delivery ↓
at macula densa
→ **MACULA DENSA**
End thick ascending limb / early distal tubule
→ **Local signalling to**
JG granular cells
→ **Renin release ↑**
→ **ANGII pathway activated**
See D4 and E

Restored NaCl delivery reduces this input. Macula densa signals; JG granular cells release renin.

D3 Renal sympathetic input

Renal sympathetic
activity ↑
→ **JG granular-cell**
β_1 receptor stimulation†
→ **Renin release ↑**
→ **ANGII pathway activated**
→ **Salt / water conservation**
+ vascular support

This is a renal branch of the sympathetic output in C3 / E5.

D4 Enzymatic cascade: renin → ANGII

JG-cell RENIN
→ **LIVER angiotensinogen**
— RENIN → angiotensin I
→ **Endothelial ACE**
Prominent in pulmonary vessels; not lung-only
→ **ANGIOTENSIN II**
→ **Four outputs:**
aldosterone / thirst / VP / vasoconstriction (E)

NEGATIVE FEEDBACK: ANGII inhibits further renin release: short-loop feedback†.

E ALL FOUR ANGII OUTPUTS

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

E1 ANGII → aldosterone

ANGII ↑
→ **ADRENAL CORTEX**
AT_1 receptor
→ **Aldosterone production**
/ secretion ↑
→ **Distal-nephron MR**
Transport proteins ↑
See F1–F2
→ **Na^+ retention ↑**
K^+ excretion ↑
Volume supported

Aldosterone is made in adrenal cortex and acts on the distal nephron.

E2 ANGII → thirst

ANGII ↑
→ **SFO / OVLT**
AT_1 receptors
→ **MnPO → cortical**
thirst circuits
→ **Drinking ↑**
→ **Water intake ↑**
Water / volume balance supported

Restored volume / perfusion reduces renin–ANGII drive (D1).

E3 ANGII → VP release

ANGII ↑
→ **SFO / OVLT AT_1**
→ MnPO
→ **MAGNOCELLULAR**
PVN / SON activated
VP synthesised here
→ **Posterior-pituitary**
VP release ↑
→ **Collecting-duct water**
conservation ↑
See B1–B2

This converges on the same VP system as A and C; repeated nodes are the same structures.

E4 ANGII → direct vascular action

ANGII ↑
→ **Vascular smooth-muscle**
AT_1 receptor
→ **Smooth-muscle**
contraction ↑
→ **Peripheral resistance ↑**
→ **BP supported**

Pressure / perfusion recovery reduces renin drive (D1); this action does not require a brain relay.

E5 ANGII → central sympathetic vasoconstriction

ANGII ↑
SFO / OVLT AT_1 activation
→ **MnPO**
→ **PARVOCELLULAR PVN**
→ RVLM
→ **Spinal sympathetic**
outflow ↑
→ **Vasoconstriction ↑**
BP supported
Renal renin branch: D3

Parvocellular PVN → RVLM is distinct from magnocellular PVN / SON → posterior pituitary.

F DISTAL ELECTROLYTES & FLUID BALANCE

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

F1 Aldosterone → Na^+ reabsorption

Aldosterone
From adrenal cortex (E1)
→ **DISTAL NEPHRON**
Intracellular MR
→ **APICAL ENaC ↑**
BASOLATERAL Na^+/K^+-ATPase ↑
→ **Na^+: lumen → cell**
→ interstitium / blood
→ **Na^+ retention ↑**
Associated water retention
ECF volume supported

Aldosterone does not insert AQP2. Water retention depends on available water / permeability.

F2 K^+–aldosterone negative-feedback loop†

Plasma K^+ ↑†
→ **ADRENAL CORTEX**
Aldosterone ↑
→ **Distal-nephron MR**
Na^+/K^+-ATPase + APICAL ROMK
→ **K^+: cell → lumen**
Urinary K^+ excretion ↑
→ **Plasma K^+**
toward normal

NEGATIVE FEEDBACK: Restored K^+ reduces direct K^+-driven aldosterone stimulation†.

ANGII also stimulates this aldosterone-sensitive K^+-secretory apparatus (E1).

F3 Body-fluid disturbance: compartment buffering

Body-fluid disturbance
→ **Plasma**
buffers change
→ **Interstitial fluid**
further buffers change
→ **Intracellular-fluid**
fluctuation dampened
→ **Stable intracellular**
environment protected

This is the lecture's compartment-buffering schematic, not an additional endocrine relay.

KEEP THESE VARIABLES SEPARATE

VP mainly regulates water permeability; aldosterone mainly regulates distal Na^+ / K^+ transport.

Plasma [Na^+] is a concentration. Total-body Na^+ strongly influences ECF volume.

AQP2 is apical; AQP3 / AQP4 provide basolateral water exit.

UT-A1 / UT-A3 transport urea. NKCC2, ENaC and ROMK transport ions.

Magnitude / timing also depend on renal perfusion, delivery, gradient and water availability.

G CONCENTRATING APPARATUS

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

G1 Countercurrent multiplication: build the NaCl gradient

Long JUXTAMEDULLARY
loops of Henle
Continuous tubular flow
→ **Descending limb**
Water-permeable
Filtrate concentrates
→ **Ascending limb**
Low water permeability
TAL APICAL NKCC2
→ **BASOLATERAL**
Na^+/K^+-ATPase
NaCl reabsorption ↑
→ **Tubular fluid diluted**
Corticomedullary NaCl gradient generated

Longer loops support a greater gradient. NKCC2 is bumetanide-sensitive; it is a cotransporter.

G2 Countercurrent exchange: preserve the gradient

Hairpin VASA RECTA
Low medullary blood flow
→ **Descending blood**
Water leaves
Solute enters
→ **Ascending blood**
Water enters
Solute leaves
→ **Solute washout limited**
Reabsorbed water removed
→ **Medullary gradient**
PRESERVED

Vasa recta preserve the gradient; they do not create it.

G3 Water reabsorption promotes passive urea exit

Collecting-duct
water reabsorption
→ **Luminal urea**
concentration ↑
→ **IMCD: APICAL UT-A1**
Urea enters cell
→ **BASOLATERAL UT-A3**
Urea exits to interstitium
→ **Inner-medullary urea ↑**
Osmotic gradient supported

Urea moves down its concentration gradient; VP increases UT-A1-mediated permeability (B2).

G4 Use the gradient to make concentrated urine

Medullary gradient
Outer: NaCl
Inner: NaCl + urea
→ **VP → collecting-duct V_2**
APICAL AQP2 available
→ **Water: lumen → AQP2**
→ cell → BASOLATERAL
AQP3 / AQP4
→ **Water returns to blood**
→ **Small volume of**
concentrated urine

Both permeability AND the osmotic gradient are required. High VP alone cannot replace a lost gradient.

H WHEN THE PATHWAY FAILS

STIMULUS / INPUT → STRUCTURES & TARGETS → EFFECT

H1 Central diabetes insipidus

PVN / SON synthesis or
posterior-pituitary release defect
→ **VP too low for need**
→ **V_2 / AQP2 stimulation**
inadequate
→ **Polyuria**
Thirst / drinking ↑
→ **Hypernatraemia if water**
replacement is inadequate

Intact thirst and access to water can keep plasma Na^+ near normal despite polyuria.

H2 Nephrogenic diabetes insipidus

Renal V_2 receptor
or AQP2 defect
→ **Kidney resistant to VP**
Hormone may be present
→ **Collecting-duct water**
conservation inadequate
→ **Polyuria + thirst**
→ **Hypernatraemia if water**
replacement is inadequate

This is a renal-response defect, distinct from low VP production / release.

H3 SIADH: inappropriate antidiuresis

CNS / drug context
Ectopic VP: small-cell lung cancer
→ **Inappropriate VP /**
renal V_2 activity
→ **Water retention ↑**
→ **Plasma Na^+ diluted ↓**
Usually hypotonic
→ **Brain-cell swelling**
Neurological symptoms if severe / rapid

Hyponatraemia is water excess relative to solute; it does not require primary total-body Na^+ depletion.

H4 Hyperaldosteronism

Conn adenoma OR
renovascular RAAS activation
→ **Aldosterone excess**
→ **Distal-nephron MR**
ENaC / Na^+/K^+-ATPase / ROMK effects ↑
→ **Na^+ retained; K^+ lost**
Intercalated-cell
H^+ secretion ↑†
→ **ECF volume / BP ↑**
Hypokalaemia
Metabolic alkalosis†

BP rises chiefly through volume expansion; serum Na^+ need not be high. K^+ loss can cause weakness / arrhythmias.

H5 Hypoaldosteronism

Adrenal failure / low-renin
state; ACE inhibitor / ARB / renin inhibitor
→ **Aldosterone drive ↓**
→ **Distal-nephron MR**
transport drive ↓
→ **Na^+ conservation ↓**
K^+ secretion ↓
Acid / NH_4^+ excretion ↓†
→ **Volume / BP may fall**
Hyperkalaemia ± low Na^+
Metabolic acidosis†

Low renin is not present in all chronic renal disease. Hyperkalaemia can cause cardiac / muscle effects.

ABBREVIATIONS & SCOPE

VP = ADH / vasopressin. PVN = paraventricular nucleus; SON = supraoptic nucleus; MnPO = median preoptic nucleus.

SFO = subfornical organ; OVLT = organum vasculosum of the lamina terminalis. These sensory CVOs detect osmotic / circulating ANGII information.

NTS = nucleus of the solitary tract; CVLM inhibits RVLM. CVLM / RVLM = caudal / rostral ventrolateral medulla. JG = juxtaglomerular.

MR = mineralocorticoid receptor; AT_1 = ANGII type-1 receptor. CD = collecting duct; TAL = thick ascending limb; ECF = extracellular fluid.

* VP–macula-densa V_1a–NO / PGE_2 is the source lecture model, not a universal prediction that VP increases net circulating renin.

† Supporting connections: JG β_1, direct K^+–aldosterone feedback, ANGII inhibition of renin, and distal acid / NH_4^+ handling.

Source: Dr J Chu, BBMS2006 (Renal — student version), PDF pp. 3–15, 19–46, plus the relevant classroom supplements. One complete overview; details selectively retained.